BBD GLUING LEMMA (GENERAL FORM, WITHOUT BOUNDEDNESS)

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Uniqueness of solutions. Cohomology of Sheaves, Lemma cohomology-lemma-uniqueness. Solutions are unique up to unique isomorphism and have vanishing negative exts.
- Finite glueing. Cohomology of Sheaves, Lemma cohomology-lemma-solution-in-finite-case. Existence for systems on a space with a finite covering.
- Glueing along increasing unions. Cohomology of Sheaves, Lemma cohomology-lemma-glueing-increasing-union. Existence along well-ordered increasing unions with compatible isomorphisms.

Source reference. Original reference 0D67: sources/stacks-chow/tags.txt, source lines 16512–16512. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0D68: sources/stacks-chow/tags.txt, source lines 16513–16513. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0D6A: sources/stacks-chow/tags.txt, source lines 16515–16515. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0D6B: sources/stacks-chow/tags.txt, source lines 16516–16516. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Theorem 0.1 (BBD gluing lemma)**.** *In Situation 0D67 assume*

(1) $X = \bigcup_{U \in \mathcal{B}} U$,
(2) *for* $U, V \in \mathcal{B}$ *we have* $U \cap V = \bigcup_{W \in \mathcal{B}, W \subset U \cap V} W$,
(3) *for any* $U \in \mathcal{B}$ *we have* $\mathrm{Ext}^i(K_U, K_U) = 0$ *for* $i < 0$.

Then there exists an object K *of* $D(\mathcal{O}_X)$ *and isomorphisms* $\rho_U : K|_U \to K_U$ *in* $D(\mathcal{O}_U)$ *for* $U \in \mathcal{B}$ *such that* $\rho_V^U \circ \rho_U|_V = \rho_V$ *for all* $V \subset U$ *with* $U, V \in \mathcal{B}$*. The pair* (K, ρ_U) *is unique up to unique isomorphism.*

Proof. A pair (K, ρ_U) is called a solution in the text above. The uniqueness follows from Lemma 0D68. If X has a finite covering by elements of $\mathcal{B}$ (for example if X is quasi-compact), then the theorem is a consequence of Lemma 0D6A. In the general

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case we argue in exactly the same manner, using transfinite induction and Lemma 0D6B.

First we use transfinite recursion to choose opens $W_\alpha \subset X$ for any ordinal α. Namely, we set $W_0 = \emptyset$. If $\alpha = \beta + 1$ is a successor, then either $W_\beta = X$ and we set $W_\alpha = X$ or $W_\beta \neq X$ and we set $W_\alpha = W_\beta \cup U_\alpha$ where $U_\alpha \in \mathcal{B}$ is not contained in W_β. If α is a limit ordinal we set $W_\alpha = \bigcup_{\beta<\alpha} W_\beta$. Then for large enough α we have $W_\alpha = X$. Observe that for every α the open W_α is a union of elements of $\mathcal{B}$. Hence if $\mathcal{B}_\alpha = \{U \in \mathcal{B}, U \subset W_\alpha\}$, then

$$S_\alpha = (\{K_U\}_{U\in\mathcal{B}_\alpha}, \{\rho_V^U\}_{V\subset U \text{ with } U,V\in\mathcal{B}_\alpha})$$

is a system as in Lemma 0D68 on the ringed space W_α.

We will show by transfinite induction that for every α the system S_α has a solution. This will prove the theorem as this system is the system given in the theorem for large α.

The case where $\alpha = \beta + 1$ is a successor ordinal. (This case was already treated in the proof of the lemma above but for clarity we repeat the argument.) Recall that $W_\alpha = W_\beta \cup U_\alpha$ for some $U_\alpha \in \mathcal{B}$ in this case. By induction hypothesis we have a solution $(K_{W_\beta}, \{\rho_U^{W_\beta}\}_{U\in\mathcal{B}_\beta})$ for the system S_β. Then we can consider the collection $\mathcal{B}_\alpha^* = \mathcal{B}_\alpha \cup \{W_\beta\}$ of opens of W_α and we see that we obtain a system $(\{K_U\}_{U\in\mathcal{B}_\alpha^*}, \{\rho_V^U\}_{V\subset U \text{ with } U,V\in\mathcal{B}_\alpha^*})$. Note that this new system also satisfies condition (3) by Lemma 0D68 applied to the solution K_{W_β}. For this system we have $W_\alpha = W_\beta \cup U_\alpha$. This reduces us to the case handled in Lemma 0D6A.

The case where α is a limit ordinal. Recall that $W_\alpha = \bigcup_{\beta<\alpha} W_\beta$ in this case. For $\beta < \alpha$ let $(K_{W_\beta}, \{\rho_U^{W_\beta}\}_{U\in\mathcal{B}_\beta})$ be the solution for S_β. For $\gamma < \beta < \alpha$ the restriction $K_{W_\beta}|_{W_\gamma}$ endowed with the maps $\rho_U^{W_\beta}$, $U \in \mathcal{B}_\gamma$ is a solution for S_γ. By uniqueness we get unique isomorphisms $\rho_{W_\gamma}^{W_\beta} : K_{W_\beta}|_{W_\gamma} \to K_{W_\gamma}$ compatible with the maps $\rho_U^{W_\beta}$ and $\rho_U^{W_\gamma}$ for $U \in \mathcal{B}_\gamma$. These maps compose in the correct manner, i.e., $\rho_{W_\delta}^{W_\gamma} \circ \rho_{W_\gamma}^{W_\beta}|_{W_\delta} = \rho_{W_\beta}^{W_\delta}$ for $\delta < \gamma < \beta < \alpha$. Thus we may apply Lemma 0D6B (note that the vanishing of negative exts is true for K_{W_β} by Lemma 0D68 applied to the solution K_{W_β}) to obtain K_{W_α} and isomorphisms

$$\rho_{W_\beta}^{W_\alpha} : K_{W_\alpha}|_{W_\beta} \longrightarrow K_{W_\beta}$$

compatible with the maps $\rho_{W_\gamma}^{W_\beta}$ for $\gamma < \beta < \alpha$.

To show that K_{W_α} is a solution we still need to construct the isomorphisms $\rho_U^{W_\alpha} : K_{W_\alpha}|_U \to K_U$ for $U \in \mathcal{B}_\alpha$ satisfying certain compatibilities. We choose $\rho_U^{W_\alpha}$ to be the unique map such that for any $\beta < \alpha$ and any $V \in \mathcal{B}_\beta$ with $V \subset U$ the diagram

$$\begin{array}{ccc} K_{W_\alpha}|_V & \xrightarrow[\rho_U^{W_\alpha}|_V]{} & K_U|_V \\ {\scriptstyle \rho_{W_\beta}^{W_\alpha}|_V}\downarrow & & \downarrow{\scriptstyle \rho_U^V} \\ K_{W_\beta} & \xrightarrow[\rho_V^{W_\beta}]{} & K_V \end{array}$$

commutes. This makes sense because

$$(\{K_V\}_{V\subset U, V\in\mathcal{B}_\beta \text{ for some } \beta<\alpha}, \{\rho_V^{V'}\}_{V\subset V' \text{ with } V,V'\subset U \text{ and } V,V'\in\mathcal{B}_\beta \text{ for some } \beta<\alpha})$$

is a system as in Lemma 0D68 on the ringed space U and because (K_U, ρ_V^U) and $(K_{W_\alpha}|_U, \rho_V^{W_\beta} \circ \rho_{W_\beta}^{W_\alpha}|_V)$ are both solutions for this system. This gives existence and uniqueness. We omit the proof that these maps satisfy the desired compatibilities (it is just bookkeeping). $\square$